Lab 1 extra slides

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Forgot your sudo password?

WSL: close your WSL terminals, open PowerShell, and run:

```
wsl -u root
passwd <your-username>
```

This drops you into a root shell with no password needed, so you can reset your own. Exit and reopen WSL normally afterward.

Apple Containers (macOS): same idea — get a root shell into the container from the host, no password required, and reset it:

```
container exec -u root <container-name> passwd <your-username>
```

Getting gdb

Debugging with `printf` works great, but gdb is worth learning too.

Check first, then install if needed:

```
gdb --version
sudo apt update && sudo apt install gdb
```

Also compile with `-g` (already in the lab `Makefiles`) so gdb can map instructions back to source lines and variable names.

Make sure you remember your sudo password — the one you set when you created your Linux user. (Instructions on how to reset it in the previous slide.)

GDB inside VS Code

Same gdb underneath, but: click the gutter to set a breakpoint, a live **Variables** panel instead of typing `print`, and a **Call Stack** panel that updates as you step.

Setup: open the Run and Debug view (sidebar icon, or Run -> Run and Debug). VS Code will prompt you to install the **C/C++** extension the first time — install it and you're ready to go.

Workflow: click left of a line number to set a breakpoint, press `F5`, then step over / into / out / continue with the floating toolbar and check the **Variables** / **Call Stack** panels.

Other cool tricks

Find out why your program crashed	`-fsanitize=address,undefined`
Visualize everything in your code	`from0.dev/tracer`